Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2026

Bachelor in Computer Applications
Course Title: Numerical Methods
Code No: CACS 252
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. Explain different types of errors in numerical computation with examples.
3. Apply Bisection Method to find a real root of $x^3 - x - 1 = 0$ correct up to three decimal places.
4. Construct Lagrange's interpolation polynomial for the data:

X	1	2	4
Y	1	4	16

5. Derive Newton's forward interpolation formula.
6. Solve the following set of linear equations using Gauss Jordan method:

$$x + y + z = 6$$
$$2x + 3y + z = 10$$
$$x + 2y + 3z = 13$$

7. Solve $\frac{dy}{dx} = x + y, y(0) = 1$ using Euler's method (h = 0.1) for x = 0.1.
8. Apply Trapezoidal rule to evaluate a definite integral.

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Explain Shooting Method for boundary value problems with algorithm.
10. Fit a straight line $y = ax + b$ to the following data using the Least Squares Method

x	1	2	3	4
y	2	5	7	8

11. Differentiate between Interpolation and approximation with their significances. Derive the formula for Simpson's (3/8) composite formula.

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2026

Bachelor in Computer Applications
Course Title: Software Engineering
Code No: CACS 253
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. What is system engineering? Explain different types of software.
3. Develop a detailed use-case diagram and description for an "Online Library Management System" for borrowing and returning books
4. Compare black-box and white-box testing with respect to fault detection capability.
5. Write a test case for login functionality of Hospital Management System.
6. Evaluate the role of re-engineering in extending software life.
7. What is validation? Explain different type of maintenance.
8. Write short notes: (Attempt any two) [2.5+2.5]
   a) ISO 9000
   b) Whitebox Testing
   c) CASE Tool

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Discuss the layered architectural model. Explain the main activities use in requirements engineering. What are the characteristics of a good Software Requirements Specification (SRS) document?
10. Explain the challenges in software project scheduling. Discuss any two project scheduling techniques in detail with examples.
11. Describe the principles of Agile Software Development. How does Extreme Programming implement these principles through its core practices?

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2026

Bachelor in Computer Applications
Course Title: Scripting Language
Code No: CACS 254
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. Write the code to create and remove new list item into Unordered list dynamically into JavaScript DOM. [5]
3. What is Ajax? Write code to check if phone is available or not into signup form using Ajax? [1+4]
4. What is JS arrow function? Explain use of forEach() and map () method in JavaScript. [1 + 4]
5. What is session, Describe how can we relate session with cookie? [1 + 4]
6. What is CMS? Describe major features of CMS. [1+4]
7. Describe different types of arrays in PHP with proper example. [5]
8. Write PHP code to upload file in PHP with rules of file type (png, jpg, gif) and file size maximum 500kb? [5]

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Write a program to perform Create and List Operation for student record with (roll, course, name, email, phone, result) database using PHP and MySQL? [5 + 5]
10. Difference between abstract class and interface. Write PHP code to create a parent class Book with properties title, author and price and a method displayBookDetails(). Also create two subclasses TextBook and ReferenceBook that inherit from Book. Add property subject in TextBook and edition in ReferenceBook. Create methods to display complete details of each book. [3+7]
11. Why do we need validation? Explain with examples which validation is best among client side and server-side validation. Write JavaScript program that use countries array to display 50 countries into tabular format with 4 countries in a row. [1+3+6]

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2026

Bachelor in Computer Applications
Course Title: Operating System
Code No: CACS 251
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

3. What is a monolithic kernel? How is it different from microkernel? Explain. [1+4]
3. What is multithreading? Explain different types of threads. [1+4]
4. Explain the different process states in detail. [5]
5. What are the necessary conditions for the deadlock? How deadlock is different from starvation? [4+1]
6. What is segmentation? How can it be used for the memory management? [1+4]
7. What are different page replacement algorithms? Explain. [5]
8. List some security problems in operating system. Explain the role of clock synchronization in distributed system. [2+3]

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. What is producer consumer problem? Explain the solutions of producer consumer problem using sleep and wakeup and compare it with the solution using semaphores. [1+4+5]
10. Consider the following resource allocation state:

	Allocation	*Max*	*Available*
	A B C D	*A B C D*	*A B C D*
P_0	0 0 1 2	0 0 1 2	1 5 2 0
P_1	1 0 0 0	1 7 5 0	
P_2	1 3 5 4	2 3 5 6	
P_3	0 6 3 2	0 6 5 2	
P_4	0 0 1 4	0 6 5 6	

Find the need matrix. Using Bankers algorithm, check whether the system is in safe state or not? If yes, find the execution order of the processes. [10]

11. What are seek time and disk access time? Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 2,150, and the previous request was at cylinder 1,805. The queue of pending requests, in FIFO order, is:
2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681
Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?
a) FCFS b) SSTF c) C-SCAN d) C-LOOK [2+8]

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2026

Bachelor in Computer Applications
Course Title: Database Management System — **Full Marks: 60**
Code No: CACS 255 — **Pass Marks: 24**
Semester: IV — **Time: 3 hours**

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. Define transaction. Describe ACID properties with example. [1+4]
3. Explain physical data independence and logical data independence. [2.5+2.5]
4. What is stored procedure? Describe statement level trigger and multi row enabled triger? [1+4]
5. Define schedule and serializability with example. How can you test the serializability? [2+3]
6. What is data model? What are its types? [1+4]
7. Define query processing and steps in query processing. [1+4]
8. Why indexing is essential in database? Differentiate dense index from sparse index with suitable example.

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Describe DDL and DML queries with example. Explain database system architecture. [3+7]
10. What is normalization? What are its advantages? Describe its types with appropriate examples. [2+2+6]
11. Consider the following schema: [3+2+2+1+2]

    Guest (Gid, Gname, Age, Gender
    STAFF (Sid, Sname, Age, Gender, Roles
    SERVICE (serId, serName, Price)
    ASSISTS (Gid, Sid) (Which staff assists which guest)
    PROVIDES (serId, Sid) (Which staff provides which service)

    Write SQL Queries:

    a) Find name of guests who are VIP.
    b) Find total number of guests assisted by staff Ram.
    c) Find name of staff whose name starts with A and ends with A.
    d) Display name of staff who provides laundary service.
    e) Give 10% discount to female guests on the price of service they receive.